Salt Taxes and Rent Control: Governments in the Market

ECON 201: Principles of Microeconomics
Lecture 20

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Governments in Markets

- Governments often intervene in markets by taxing goods or setting price controls.
- One reason for taxes is to raise revenue to pay for public services.
- Governments also intervene to achieve social goals. For example:
  - Progressive taxation to reduce inequality.
  - Rent control to make housing more affordable.
  - Carbon taxes to reduce emissions, or cigarette taxes to reduce smoking.

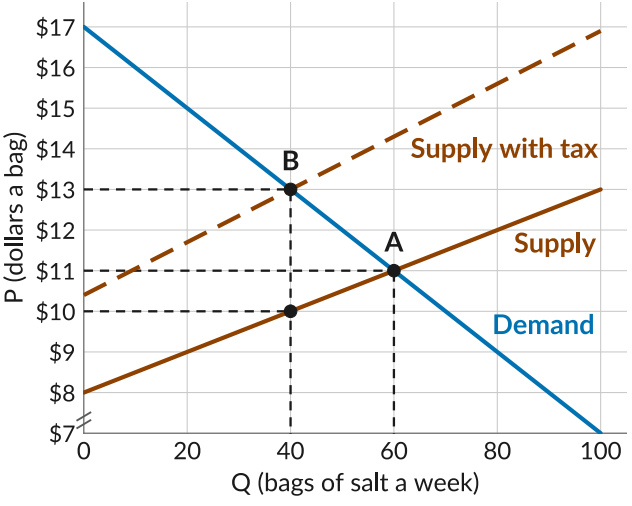
Today: we study these interventions using the supply and demand framework.

A Tax on Salt

- Salt is one of the oldest goods to be taxed. As a preservative, it allowed food to be stored, transported, and traded.
- The ancient Chinese advocated taxing salt to raise revenue because people needed it, an idea we will come back to.
- Ruling elites in ancient India, the French monarchy, Tsarist Russia, and the British in India all taxed salt.
- We will use a stylized salt market with a 30% sales tax.

A 30% Salt Tax

- Original equilibrium at A: 60 bags at $11.
- The tax raises sellers' marginal cost of each bag by 30%, so the supply curve moves up and gets steeper.
- New equilibrium at B: 40 bags at $13.
- **Note**: $13 is the price buyers pay, sellers keep only $10, and the $3 difference goes to the city.



Who Pays the Tax?

Outcome	Without tax	With tax
Buyers pay	\$11	\$13
Sellers get	\$11	\$10
Government gets	\$0	\$3
Bags sold	60	40

- Buyers pay \$2 more and sellers keep \$1 less, which adds up to the \$3 tax.
- **Tax incidence** is how a tax is shared between buyers and sellers:

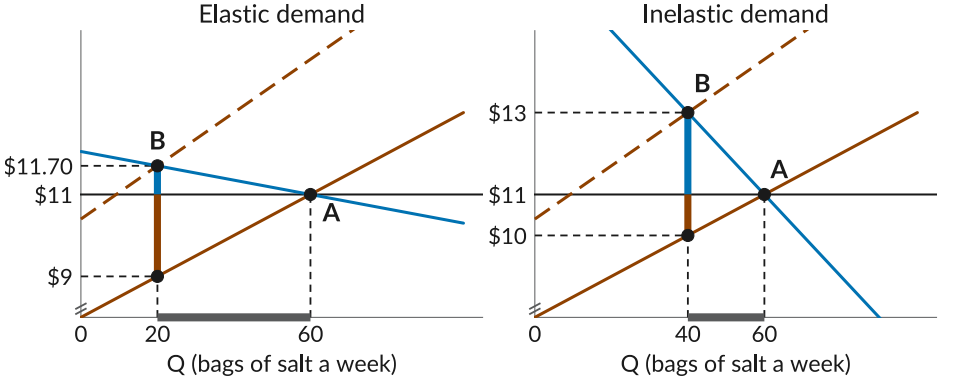
$$\text{Buyers' share} = \frac{2}{3} = 67\% \quad \text{and} \quad \text{Sellers' share} = \frac{1}{3} = 33\%$$

What Determines Who Bears the Tax?

- **Key insight**: tax incidence, how much of the tax each side bears, does not depend on who is taxed.
  - The sales tax is legally imposed on sellers, but that does not tell us who bears its burden.
- Then what determines who bears the burden of the tax? The less **elastic side** of the market bears more.
  - If buyers need salt and have few alternatives, they will buy even at a higher price, so sellers can pass on much of the tax.

Tax Incidence and Elasticity

The same 30% tax, with two demand curves through A.



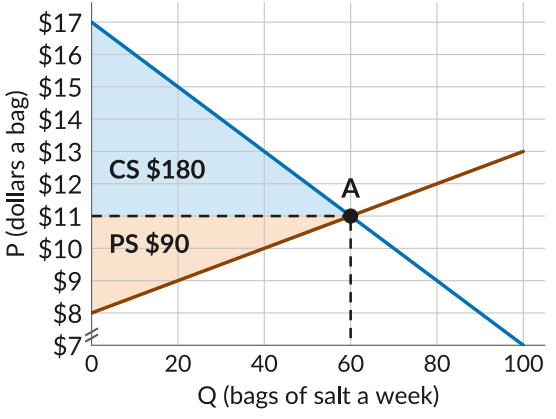
Buyers: $0.70 more. Sellers: $2 less.

Buyers: $2 more. Sellers: $1 less.

The less elastic side of the market bears more of the tax.

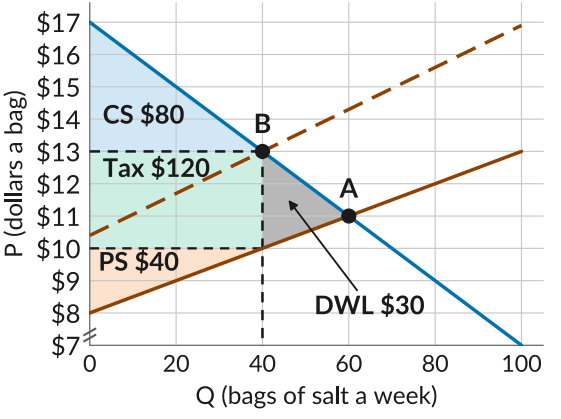
Surplus, Revenue, and Deadweight Loss

Without the tax



Total surplus = $180 + $90 = $270

With the tax



Total surplus = $80 + $40 + $120 = $240

What Makes a Good Tax?

- **To raise as much revenue as possible**, and reduce deadweight loss, the government would prefer to tax a good with low elasticity of demand, so that the fall in quantity is quite small.
- But this also means that the incidence of the tax falls heavily on consumers.
- The French monarchy's salt tax was so resented it helped bring on the French Revolution, Gandhi's 1930 salt march challenged British India's salt tax, and the 1773 Boston Tea Party protested a British tax on tea.
- **To cut consumption** of something harmful, like tobacco or carbon, a tax works best when demand is elastic, so the quantity falls a lot.

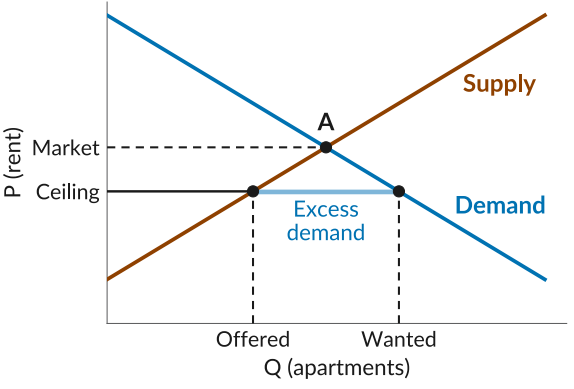
Price Controls

- Another lever governments use is price controls: legal limits on a price.
- A **price ceiling** is a maximum price, set below the market level.
  - Examples: rent control, capped taxi fares, price gouging laws, and caps on interest rates.
- A **price floor** is a minimum price, set above the market level.
  - Examples: the minimum wage, farm price supports, and minimum alcohol prices.
- In Lecture 17, we saw that a price above or below the equilibrium creates a **deadweight loss**. We can now apply that framework directly here.

Rent Ceilings

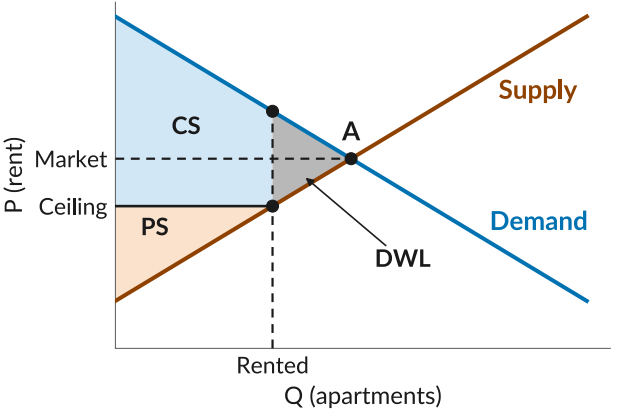
Rent ceiling: the maximum rent a landlord can legally charge, set below the market level for goals such as fairness.

- Without a ceiling, the market clears at A.
- At the ceiling, renters want more apartments than landlords offer: an **excess demand**.
- Trade happens on the **short side**, the side that wants to trade less: here, landlords.



Who Gains From a Rent Ceiling?

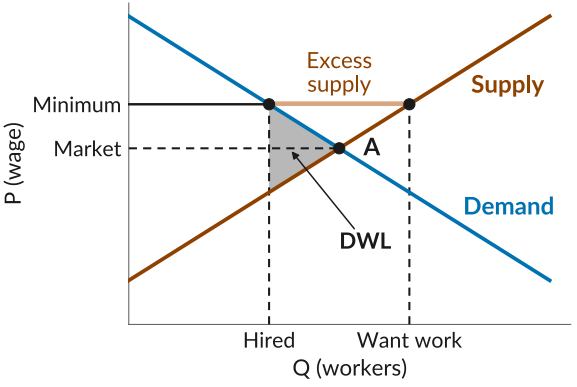
- Renters who get an apartment gain from the lower rent, while landlords lose: fewer apartments at a lower rent.
- The apartments no longer rented were worth more than they cost: a **deadweight loss**.
- The apartments may not go to those who value them most.



In the long run: building more housing shifts the supply curve out, which can close the excess demand.

The Minimum Wage

- The **minimum wage** is a price floor on labor.
- Set above the market wage, it creates a **deadweight loss**: firms hire fewer workers, and some who want to work cannot find jobs.
- In many real-world labor markets, it is below the market wage, so it has no effect.
- Also, many labor markets are not perfectly competitive: employers have some market power.



Are Taxes and Price Controls Always Costly?

- We saw that both taxes and price controls, even when aimed at good goals, create deadweight losses.
- If the tax revenue is spent on goods and services that improve people's wellbeing, that benefit to society may outweigh the surplus lost in the taxed market.
- Again remember, that these losses hold only if the conditions for a competitive equilibrium to be efficient are met, and the real world is less than perfect.
- Before Midterm 1, we saw markets that deviate from perfect competition, and after this midterm we turn to more of them. In these markets, government interventions like these can make sense again.

Midterm 2

- **Midterm 2 is Monday, November 9**, and covers Lectures 15 to 20.
- The Exams page has the **Module IV notes**, the **study guide**, and two **practice exams** with solutions.
- To prepare, read the notes, go through the study guide, do the practice problems, and finish with the practice exams, timed.
- Bring a basic calculator. There is no help sheet.